

Kaiwen Bian

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Education

- 2022 – 2026 **University of California, San Diego**, La Jolla, CA
B.S. in Data Science, Major-GPA 3.891/4.0
B.S. in Cognitive Behavioral Neuroscience, Major-GPA 3.875/4.0
Research Interests: Embodied Agent, World Modeling, Deep Representation Learning, AI4Science, Foundation Model.

Research Experience

- 2026 – Now **Stanford University**, *Staff Researcher*, Palo Alto, CA.
Enigma Project, Department of Ophthalmology & Stanford Bio-X
Advisor: Andreas S Tolias
 - Building brain foundation models and harvesting natural intelligence to build less artificial intelligence.
- 2024 – 2026 **Salk Institute for Biological Studies**, *Research Intern*, La Jolla, CA.
Crick-Jacobs Center for Theoretical and Computational Biology
Advisor: Talmo D Pereira
 - Developed computationally efficient deep reinforcement imitation systems (Mimic-MJX) to control a bio-mechanically realistic agent and mimic realistic animal behaviors.
 - Discovered governing topological insights into the representational space of embodied agents' behaviors (TopoMimic) and achieved higher performance on behavior transition clustering compared to SOTA segmentation methods.
 - Built a switching nonlinear dynamical system (Code2Act) that maps discrete behavioral syllables to continuous control through learned internal dynamics.
- 2025 – 2026 **Stanford University**, *Visiting AI Researcher*, Palo Alto, CA.
Statistics Department & Wu Tsai Neurosciences Institute
Advisor: Scott W Linderman
 - Created latent dynamical models for planning and generating bio-mechanically realistic behaviors for embodied agents using deep state-space modeling methods.
 - Co-advised on the Code2Act project from Salk.
- 2025 – 2026 **University of California, San Diego**, *Student Researcher*, La Jolla, CA.
Halicioğlu Data Science Institute
Advisor: Yusu Wang, Gal Mishne
 - Created the Graph Compositional Abstraction Scheme for Tokenization (MOSAIC), allowing sequence transformers to generate complex molecules more realistically with more structural properties compared to SOTA flat tokenization methods.
 - Co-advised on the TopoMimic project from Salk.

2023 – 2024 **University of California, San Diego, FMP Scholar**, La Jolla, CA.

Cognitive Science Department & Undergraduate Research Hub

Advisor: Sean Trott

- Developed tools using linguistic techniques (i.e. affordance) for probing the embodied simulation in large multi-modal models to improve the interpretability of these models.

Conference & Journal Publications

- Nature 2026 Zhang, C.Y., Yang, Y., Sirbu, A., Leonardis, E.J., Abe, E., Warnberg, E., Aldarondo, D.E., Lee, A., Prasad, A., Foat, J., **Bian, K.**, Park, J., Bhatt, R., Saunders, H., Nagamori, A., Thanawalla, A.R., Huang, K.W., Plum, F., Beck, H., Flavell, S.W., Labonte, D., Richards, B.A., Brunton, B.W., Azim, E., Ölveczky, B.P., & Pereira, T.D. (2026). **MIMIC-MJX: Neuromechanical Emulation of Animal Behavior**. Manuscript submitted to *Nature Methods*.
- NeurIPS 2026 **Bian, K.**, Yang, A. H., Parviz, Ali., Mishne, G., & Wang, Y. (2026). **Beyond Flat Walks: Compositional Abstraction for Autoregressive Graph Generation**. Manuscript submitted to *Advances in Neural Information Processing Systems*.
- PMLR 2026 **Bian, K.**, Leonardis, E. J., Yang, Y., Zhang, C., Wang, Y., & Pereira, T. D. (2026). **Shaped by What's Missing: Topological Invariance Simplification Discovers Behavioral Transitions**. Manuscript submitted to *Advances in Neural Information Processing Systems Workshop: Symmetry and Geometry in Neural Representations*.
- NeurIPS 2026 **Bian, K.**, Zhang, C. Y., Yang, Y., Sirbu, A., Jha, A., Buchanan, K., Leonardis, E. J., Richards, B. A., Ölveczky, B. P., Linderman, S. W., & Pereira, T. D. (2026). **Code to Action: Motor Programs from Discrete Syllables in Whole-Body Biomechanical Control**. Manuscript in preparation to submit to *Advances in Neural Information Processing Systems Workshop: Neuro-Symbolic Embodied Intelligence*.

Conference Presentations

- SimBodies 2026 **Bian, K.**, Zhang, C. Y., Yang, Y., Sirbu, A., Jha, A., Buchanan, K., Leonardis, E. J., Richards, B. A., Ölveczky, B. P., Linderman, S. W., & Pereira, T. D. (2026). **Code to Action: Motor Programs from Discrete Syllables in Whole-Body Biomechanical Control**. Accepted at *Simulated Bodies: Whole Body Biomechanical Models (SimBodies) Conference*, Lansdowne, VA.
- UCSD 2026 **Bian, K.**, Yang, A. H., Parviz, Ali., Mishne, G., & Wang, Y. (2026). **Beyond Flat Walks: Compositional Abstraction for Autoregressive Molecular Generation**. Accepted at *Halicioğlu Data Science Institute, UC San Diego Senior Capstone Showcase*, La Jolla, CA.

- COSYNE 2026 Zhang, C., Sirbu, A., Yang, Y., Leonardis, E. J., Park, J., Prasad, A., **Bian, K.**, Abe, E., Wörnberg, E., Brunton, B. W., Richards, B., Ölveczky, B. P., & Pereira, T. D. (2026). **MIMIC-MJX: Neuromechanical Emulation of Animal Behavior**. *Poster will be presented at Computational and Systems Neuroscience (Cosyne) Conference, Lisbon, Portugal.*
- COSYNE 2026 Leonardis, E. J., Nagamori, A., Thanawalla, A., Yang, Y., Saunders, H., Gilmer, J., Zhang, C., **Bian, K.**, Ölveczky, B. P., Al Borno, M., Azim, E., & Pereira, T. D. (2026). **Musculoskeletal Imitation Learning: Physics-Aware Constraints Promote Naturalistic Muscle Activity**. *Accepted at Computational and Systems Neuroscience (Cosyne) Conference, Lisbon, Portugal.*
- Stanford 2025 **Bian, K.**, Jha, A., Buchanan, K., & Linderman, S. W. (2025). **Deep State Space Controls For Biomechanically Realistic Artificial Agents**. *Poster presented at Stanford Undergraduate Research Program (SURP-Stats) Symposium, Palo Alto, CA.*
- SfN 2025 **Bian, K.**, Leonardis, E. J., Yang, Y., Zhang, C., Azim, E., Ölveczky, B. P., Wang, Y., & Pereira, T. D. (2025). **Topology-driven Insights into Naturalistic Behavior from Neuromechanical Agent Modeling**. *Poster presented at the Society for Neuroscience (SfN) Annual Meeting, San Diego, CA.*
- SfN 2025 Yang, Y., Zhang, C., Leonardis, E. J., Sirbu, A., **Bian, K.**, Azim, E., Ölveczky, B. P., & Pereira, T. D. (2025). **VNL-playground: Fast and Biologically Realistic Virtual Environment for Simulating Animal Behavior**. *Poster presented at the Society for Neuroscience (SfN) Annual Meeting, San Diego, CA.*
- SfN 2025 Leonardis, E. J., Nagamori, A., Thanawalla, A., Yang, Y., Saunders, H., Gilmer, J., Zhang, C., **Bian, K.**, Ölveczky, B. P., Al Borno, M., Azim, E., & Pereira, T. D. (2026). **Musculoskeletal Imitation Learning: Physics-Aware Constraints Promote Naturalistic Muscle Activity**. *Poster presented at the Society for Neuroscience (SfN) Annual Meeting, San Diego, CA.*
- COSYNE 2025 Zhang, C., Yang, S., **Bian, K.**, Abe, E., Wörnberg, E., Foat, J., Aldarondo, D., Brunton, B. W., Ölveczky, B. P., & Pereira, T. D. (2025). **Track-MJX: A GPU-Accelerated Pipeline for Imitating Animal Motor Control**. *Poster presented at the Computational and Systems Neuroscience (COSYNE) Conference, Montreal, Canada.*
- UCSD 2023 **Bian, K.**, Li, A., Jones, C., & Trott, S. (2023). **Embodied Simulation in Multimodal Models Using Affordance Stimulus: A Probing Study**. *Poster presented at the UC San Diego Undergraduate Research Faculty Mentorship Program (FMP) Symposium, La Jolla, CA.*

Skills

Programming Python, SQL, Java, Java Script, Web Dev.

Python Packages Jax, Flax, Ray, MuJoCo/MJX, Gym, Dm-Control, Brax, PyTorch (Torch, TorchRL, and PyG), TensorFlow, Ripser, SkLearn, Panda, Dask, and Spark.

Mathematic Linear Algebra, **Machine/Deep Learning, Reinforcement Learning, Geometric Learning** (Graph Learning, Computational Topology, and Manifold Learning), Probability/Statistics (State Space Models, Probabilistic Inference, Stochastic Processes, and Computational Stochastic), Convex Optimization, Dynamical System, Signal Processing, and Linear Control Theory.

Neuroscience & Biology System & Cognitive Neuroscience, Neuro-anatomy, Motivational & Developmental Neurobiology, Behavioral Endocrinology, Neural Signal Processing, Exercise Physiology, Kinesiology, and Human Nutrition.

Language Mandarin Chinese and English